HOTEL BOOKING

CANCELLATION PREDICTION

A machine-learning approach to understanding booking cancellations

PROJECT PRESENTATION

Classification • Model comparison • PCA experiment

The business problem

Can we identify bookings likely to be cancelled early enough to support better hotel planning?

- Target: is_canceled (0 = not canceled, 1 = canceled)
- Task: binary classification
- Potential value: support room planning, staffing and revenue decisions
- Example features: lead time, hotel type, deposit type and special requests

DATA

Hotel Booking Demand dataset

PREDICTION

Estimate cancellation risk for each booking

DECISION SUPPORT

Flag higher-risk bookings for review

Prepare reliable data

Cleaning choices focus on data quality and preventing information leakage.

DUPLICATES

Remove duplicate rows so repeated records do not distort analysis.

MISSING VALUES

Inspect missingness; impute within the training pipeline.

OUTLIERS

Use IQR to flag unusual values; keep plausible bookings.

DATA TYPES

Treat agent and country as categorical variables where appropriate.

LEAKAGE PREVENTION

Drop reservation_status and reservation_status_date because they reveal the outcome.

COMPANY COLUMN

Dropped in this workflow due to extensive missingness and limited usefulness.

Outliers are not automatically errors; unusual prices or lead times may be genuine bookings.

Explore patterns before modelling

TARGET DISTRIBUTION

Compare counts for canceled vs. non-canceled bookings.

HOTEL TYPE

Compare cancellations between City Hotel and Resort Hotel.

LEAD TIME

Compare lead-time distributions across target classes.

DAILY RATE (ADR)

Compare room-rate distributions for both outcomes.

CORRELATION HEATMAP

Explore linear relationships between numerical features.

Takeaway: charts help form hypotheses; the test set checks whether predictions generalise.

Preprocessing without data leakage

1 → 2 → 3 → 4 → 5

SPLIT

80% train
20% test

IMPUTE

Median / most frequent

ENCODE

One-hot encode categories

SCALE

Standardise numeric values

FIT

Learn transforms on train only

Pipelines keep preprocessing consistent and prevent test data from influencing learned imputers, encoders or scalers.

Compare three classification models

Model	Test accuracy	Precision	Recall	Test F1	ROC-AUC
Logistic Regression	0.750	0.531	0.794	0.636	0.845
Decision Tree	0.799	0.596	0.830	0.694	0.879
Random Forest	**0.793**	**0.585**	**0.857**	**0.695**	**0.898**

Best F1

RANDOM FOREST

Test F1 = 0.695

RANDOM FOREST

Recall = 0.857

DECISION TREE

Accuracy = 0.799

Use the same split for all models. Re-run the notebook to confirm values match the final cleaned dataset.

Choose a model for the business need

PRECISION

When the model flags a cancellation, how often is it correct?

RECALL

Of all actual cancellations, how many does it identify?

F1 SCORE

Balances precision and recall when both errors matter.

Selected model: Random Forest

- Highest recorded test F1: 0.695
- Highest recall: 0.857 — identifies more actual cancellations
- Highest ROC-AUC: 0.898 in this comparison
- Decision Tree had slightly higher accuracy: 0.799 vs 0.793

PCA: useful experiment, not automatic improvement

PREPROCESS
Impute • encode • scale

→

PCA
100 components

→

RANDOM FOREST
Predict cancellation

WITHOUT PCA
Random Forest test F1
0.695

WITH PCA
Random Forest test F1
0.688

RESULT
PCA was slightly lower on F1.

PCA compresses features into components, but those components can be harder to interpret and may not help tree-based models.

Conclusion & next steps

WHAT WORKED

Consistent preprocessing and a fair comparison of three classifiers.

BEST CURRENT MODEL

Random Forest based on test F1, recall and ROC-AUC.

PCA FINDING

PCA did not improve F1 in the recorded comparison.

Business impact

The model can help flag bookings with higher cancellation risk and support planning. Predictions should inform decisions, not replace them.

Limitations & next steps

- Validate with cross-validation
- Tune hyperparameters and decision threshold
- Check performance across hotel types and time periods
- Monitor model drift over time

THANK YOU

Questions?

Hotel Booking Cancellation Prediction

